Sivateja Varma Jampana

Software Engineer

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Professional Summary

Software Engineer with a Bachelor of Technology in Computer Science and specialized expertise in Python and FastAPI. Experienced in architecting scalable RESTful APIs, microservices architecture, and cloud deployments. Skilled in writing clean, efficient, and well-documented backend code while integrating modern frontend interfaces, Docker containerization, and AWS infrastructure.

Work Experience

Software Developer Intern May 2025 – October 2025
GYT Workz *Hyderabad, India*

- Engineered a real-time computer vision processing application in Python using OpenCV to evaluate drone-captured video streams.
- Optimized frame processing pipelines using grayscale conversion and adaptive thresholding algorithms, reducing execution latency by 35% and boosting processing efficiency.

Education

Bachelor of Technology in Computer Science and Engineering Aug. 2021 – May 2025
Gandhi Institute of Technology and Management (GITAM) *Visakhapatnam, India*

Extracurricular Activities

Placement Coordinator & Marketing Team Member Aug. 2022 – Present
Career Guidance Center (GCGC) & ISTE Group *Visakhapatnam, India*

Technical Skills

Languages & Frontend: Python, JavaScript, HTML, CSS
Frameworks & Libraries: FastAPI, React, OpenCV
Databases & Systems: PostgreSQL, MySQL, Redis, Redis Streams
Cloud & DevOps: AWS (EC2), Docker (Containerization), Git (Version Control), Railway
Core Competencies: RESTful API Design, Microservices, Asynchronous Processing, JWT Authentication, Unit Testing, Debugging

Projects

E-Commerce Microservices Platform | *Python, FastAPI, Redis, Redis Streams, React* 2026

- Architected an event-driven microservices backend decoupling Payment and Inventory domains using FastAPI, Redis, and Redis Streams.
- Implemented asynchronous, cross-service communication utilizing Redis Streams and Consumer Groups, ensuring 100% message delivery reliability and eliminating duplicate order processing.
- Engineered a resilient fault-tolerance mechanism employing compensating transactions to automatically trigger stream-based refund workflows upon inventory allocation failures.
- Accelerated API processing throughput by executing non-blocking payment tasks via FastAPI Background Tasks, reducing latency by 40%.
- Developed a responsive frontend interface using React to communicate seamlessly with decoupled microservices via RESTful APIs.

Car API | *http://32.236.44.143/* | *FastAPI, React, PostgreSQL, Docker, AWS* May 2026

- Designed, developed, and deployed a public Python RESTful API developer tool, managing the full Software Development Life Cycle (SDLC) from architecture to production.
- Integrated React frontend components with a scalable backend architecture supporting PostgreSQL and secure JWT-based authentication.
- Managed codebase version control using Git and containerized the application with Docker for seamless continuous deployment on an AWS EC2 instance.